TECHNICAL PRESENTATION

Encryption and Decryption Using AES &RSA

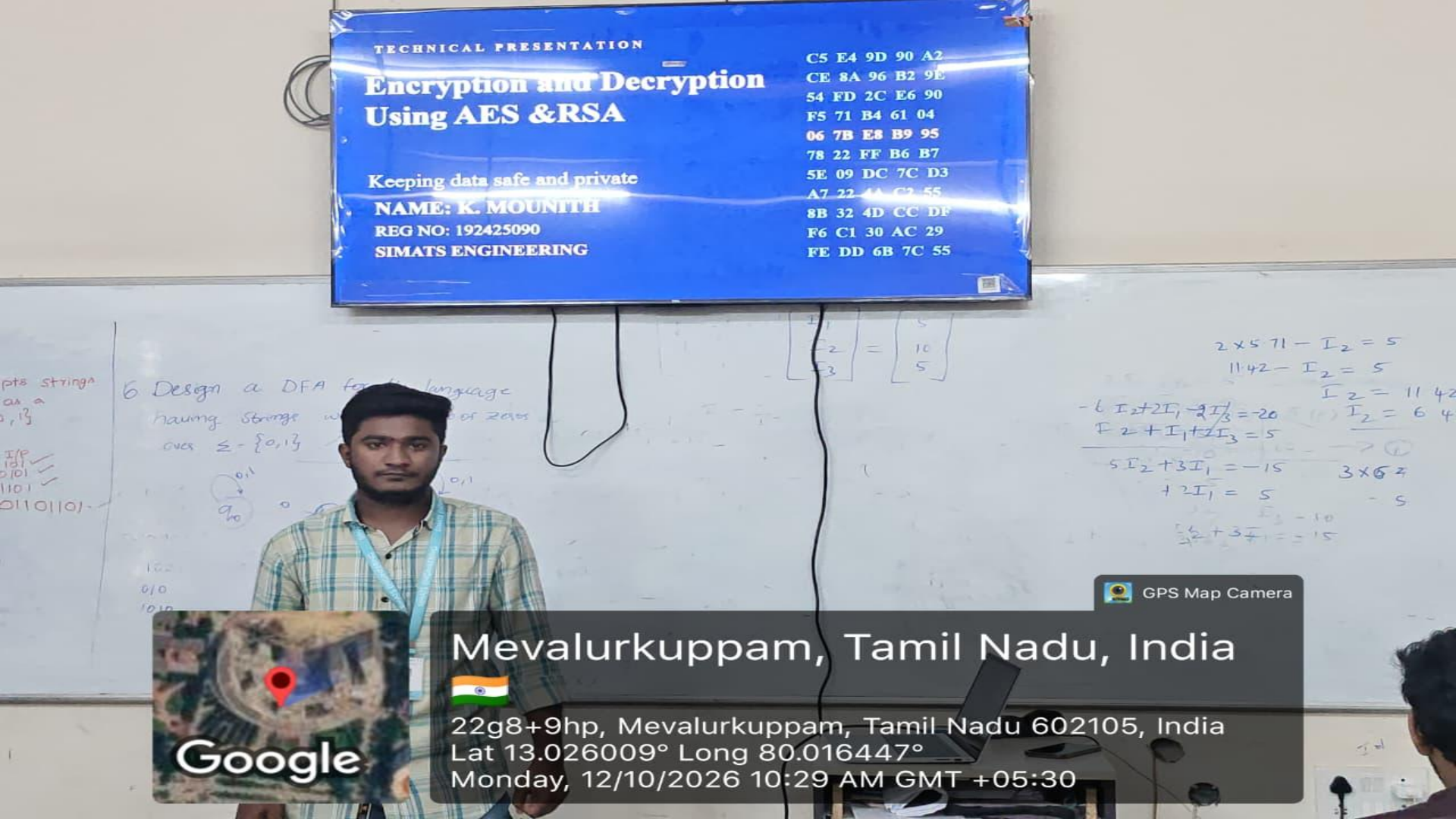
Keeping data safe and private

NAME: K.MOUNITH

REG NO: 192425090

SIMATS ENGINEERING

C5 E4 9D 90 A2
CE 8A 96 B2 9E
54 FD 2C E6 90
F5 71 B4 61 04
06 7B E8 B9 95
78 22 FF B6 B7
5E 09 DC 7C D3
A7 22 4A C2 55
8B 32 4D CC DF
F6 C1 30 AC 29
FE DD 6B 7C 55


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GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
22g8+9hp, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026009° Long 80.016447°
Monday, 12/10/2026 10:29 AM GMT +05:30
Google

What Are Encryption and Decryption?

LOCK

Encryption

Changing readable data into a secret code

UNLOCK

Decryption

Changing the secret code back into readable data

A secret key is used in both steps.

How It Works



Plain text = readable data. Cipher text = secret code.

1

Sender locks the data with a key

2

Data travels as secret code

3

Receiver unlocks it with a key

Why Do We Need It?

Data travels across networks where others may see it. Encryption keeps it safe.

1. **Privacy**
   Keeps our data secret
2. **Security**
   Stops hackers reading it
3. **Trust**
   Makes online use safe

Two Types of Encryption

1 Symmetric

One key

- Same key locks and unlocks
- Fast and simple
- Key must be shared secretly

2 Asymmetric

Two keys

- Public key locks the data
- Private key unlocks it
- Safer to share, but slower

Symmetric vs Asymmetric

	Symmetric	Asymmetric
Keys	One key	Two keys
Speed	Fast	Slower
Key sharing	Difficult	Easy
Best for	Large data	Secure key exchange

Common Algorithms

AES

Symmetric algorithm

Protects files and data. Very fast and widely trusted.

RSA

Asymmetric algorithm

Shares keys safely and signs data using a public and a private key.

Where Is It Used?

1. **Messaging apps**

   Keeps chats private

2. **Online banking**

   Protects payments

3. **Websites (HTTPS)**

   Makes browsing secure

4. **Stored files**

   Protects data on devices

Threats and Safety Tips

Common Threats

- Hackers guessing keys
- Weak or stolen passwords
- Fake websites

Safety Tips

- Use strong keys and passwords
- Visit HTTPS websites
- Keep software updated

Conclusion

- Encryption turns data into secret code
- Decryption turns it back using the key
- Together they keep our data safe and private

Thank you